Research Techniques

BY

PRADIP PAUDYAL, PHD

What will you be studying?

Specific Objectives	Contents
Organize a scientific research process to solve a real world problem.	**1. Introduction to Research (5 hrs)** 1. Research: Objectives and Characteristics 2. Types of Research 3. The 6 P's of Research 4. Research Process 5. Research in Computer Science and Engineering
Apply literature review and identify a research gap to formulate a problem.	**1. Literature Review (6 hrs)** 1. Purpose of Literature Review 2. Literature Resources and Techniques 3. Conducting a literature review 4. Referencing and Citation 5. Evaluating literature review 6. Identifying problem: Research question / problem statements and hypothesis 7. Case Study: Prepare a review paper.
Apply the research design and strategies based on the nature of the problem chosen to solve.	**3. Research Methodology (8 hrs.)** 1. Components of Research Methodology: Research problem, Objectives, Research Design, Data Collection, Sampling Techniques, Data Analysis, Hypothesis Formulation, Ethical Consideration and Validation and Reliability 2. Research Design: Experimental, Exploratory, Descriptive and Explanatory 3. Research Strategies: Experimental Research, Case Study, Action Research, Theoretical Research, Simulation-based Research. 4. Multidisciplinary Research Approaches
Use appropriate data collect methods to generate the data required for the chosen problem and evaluate and validate the data.	**4. Data Collection (6 hrs.)** 1. Primary and Secondary Data 2. Data Collection Methods: Surveys, Experiments, Questionnaires, Interviews, Observation, Experiments, Documents 3. Data Preparation 4. Data Sampling and Its Techniques 5. Evaluation and Validation of Data

What will you be studying?

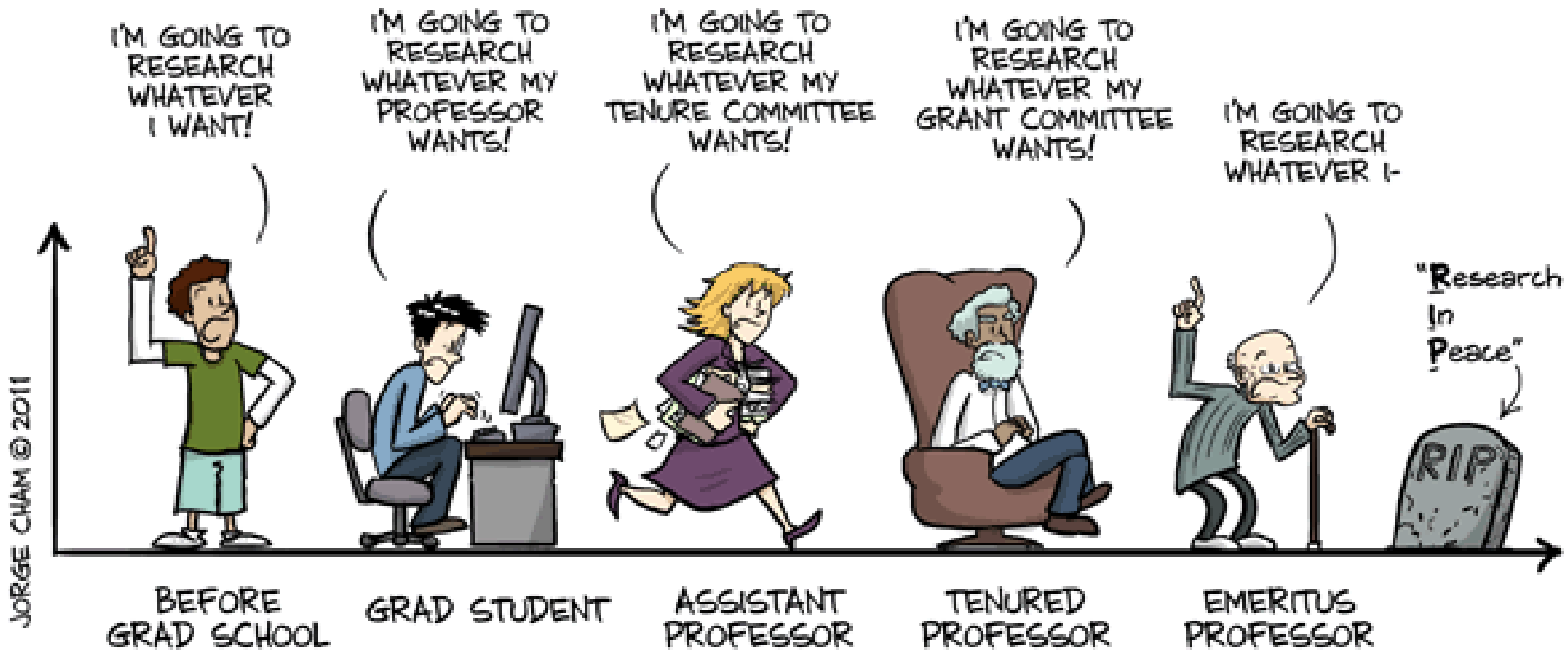
Apply the data analysis and visualization tools / techniques to analyze and visualize the pattern in the data.	**5. Data Analysis (12 hr)** 1. Quantitative Data Analysis: Types of Quantitative Data, Data Coding, Data Presentation: Tables, Bar charts, Scatter Graph, Line graph 2. Analysis Techniques: Descriptive Statistics, Inferential Statistics (Z-test, t-test, Chi-square test, ANOVA), Regression Analysis, Time Series Analysis and Multivariate Analysis 3. Qualitative Data Analysis: Qualitative Data, Analyzing Textual Data, Analyzing Non-Textual Data 4. A detailed Case Study in data visualization and analysis using Python libraries: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Scikit-learn
Apply the research ethics during the entire research work.	**5. Ethical Issues in Research (4 hrs)** 1. Right of People directly involved 2. Responsibilities of an Ethical Researcher 3. Data Privacy and Security 4. Plagiarism and Intellectual Property 5. Algorithmic Bias and Fairness 6. Environmental Responsibility 7. Conflict of Interest 8. Copyright and Patent 9. ACM Code of Ethics and IEEE Code of Ethics
Write the research work and present it in research papers, thesis, conference and journal.	**5. Presentation of the Research (4 hrs)** 1. Writing up the research 2. Written Presentation: Academic papers, Theses and dissertations, Technical reports 3. Visual Presentations: Posters, Infographics, Demonstration 4. Oral Presentation: Conference talks, Thesis defense 5. Conferences and Journal **[Students must present (oral) a review paper or any research paper prepared by them in class or national/international conference]**

Oates, B. J., Griffiths, M., & McLean, R. (2022). *Researching information systems and computing.* Sage..

About Research

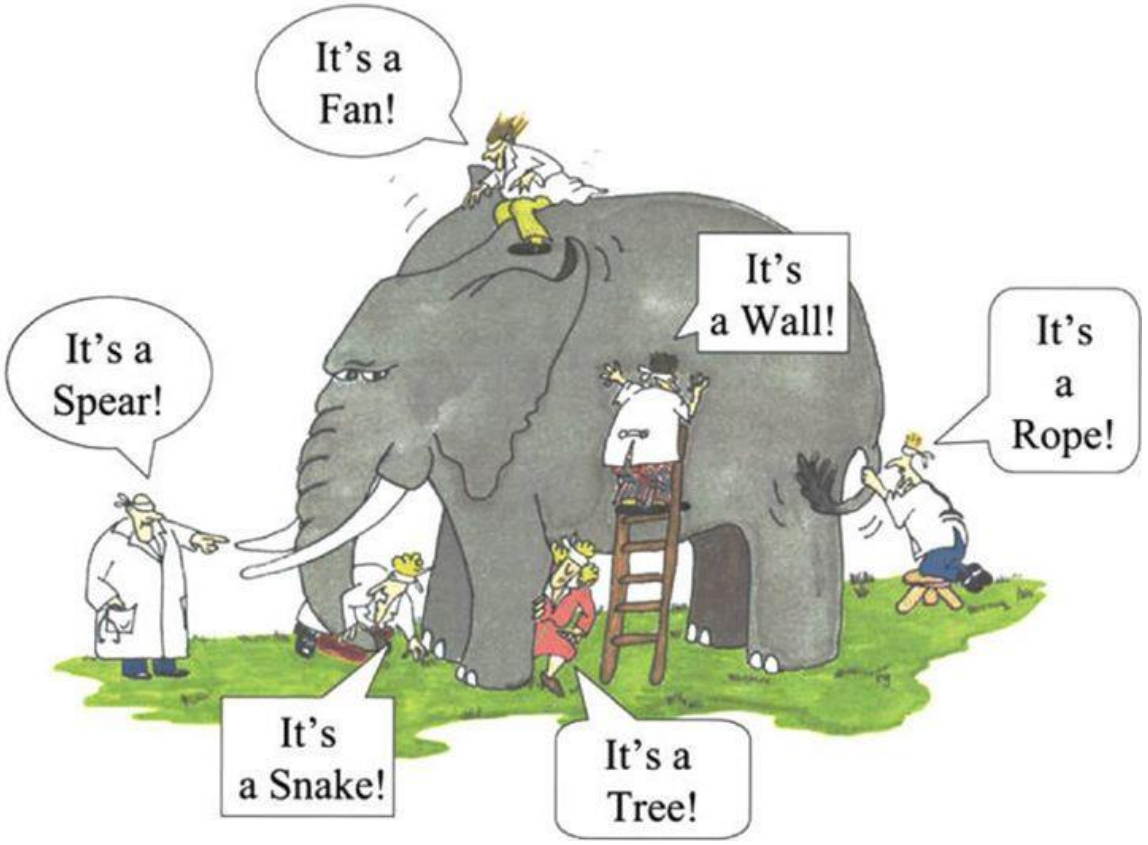


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What is Research?



What is Research?

Here are some widely accepted definitions of **research** from different perspectives:

1. **General Definition:**
   - **"Research is a systematic inquiry to describe, explain, predict, and control observed phenomena."** – Creswell
   - **"Research is the systematic process of collecting and analyzing information to increase our understanding of a topic or issue."** – Leedy and Ormrod
2. **Academic Definition:**
   - **"Research is a careful investigation or inquiry, especially through search for new facts in any branch of knowledge."** – The Advanced Learner's Dictionary
   - **"Research comprises creative work undertaken systematically to increase the stock of knowledge, including knowledge of humanity, culture, and society, and the use of this stock of knowledge to devise new applications."** – OECD
3. **Business Context:**
   - **"Research is a systematic approach to problem-solving, aiming to understand customer needs, market trends, and other factors that drive business decisions."**
4. **Scientific Perspective:**
   - **"Research is the process of addressing a scientific question through experimentation, observation, and analysis."**

Each definition emphasizes the **structured approach** and the goal of advancing knowledge. Would

What is Research?

According to Rusk

"Research is a point of view, an attitude of inquiry or a frame of mind. It asks questions which have not been asked, and it seeks to answer them by following a fairly definite procedure. It is not a mere theorizing, but rather an attempt to elicit facts and to face them once they have been assembled."

According to John W. Best

"Research is considered to be the more formal, systematic, intensive process of carrying on the scientific methods of analysis. It involves a more systematic structure of investigation, usually resulting in some sort of formal record of procedures and a report of results or conclusions."

According to C. Francies Rummel

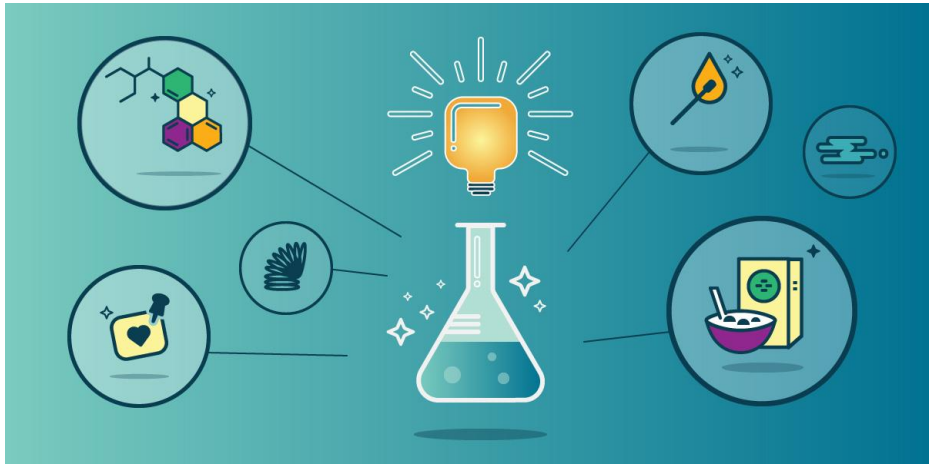
"Research is an endeavour to discover, develop and verify knowledge. It is an intellectual process that has developed over hundreds of years, ever changing in purpose and form and always searching for truth."

"… systematic investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge."

Research is not

✓**Accidental discovery**

- Accidental discovery may occur in structured research process
- Usually takes the form of a phenomenon not previously noticed
- May lead to a structured research process to verify or understand the observation



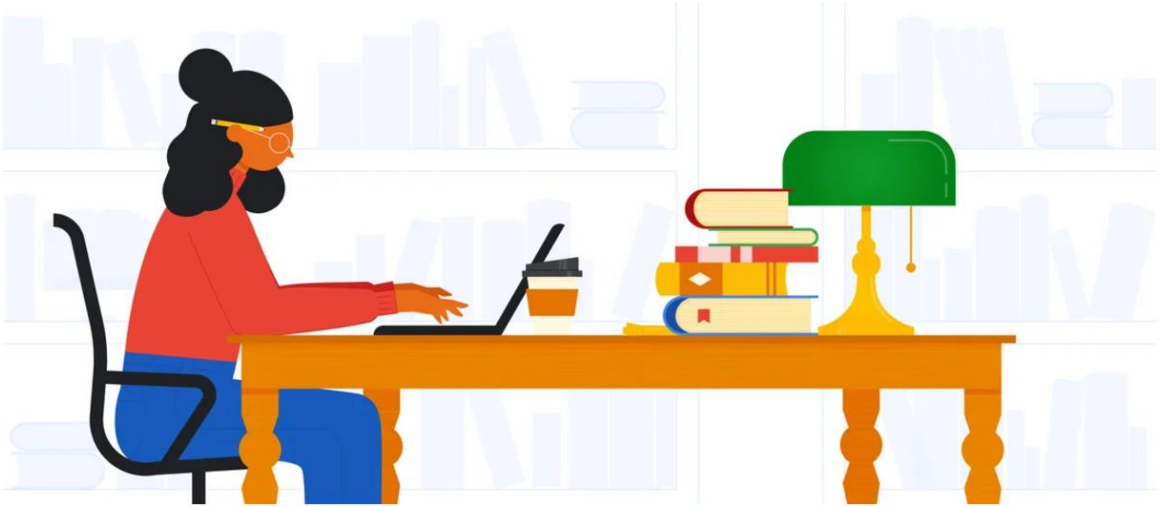
Research is not

- ✓**Data Collection**
  - an intermediate step to gain reliable knowledge
  - collecting reliable data is part of the research process



Research is not

- ✓**Searching out published research results in libraries** (or the internet)
  - This is an important early step of research
  - The research process always includes synthesis and analysis
  - But, just reviewing of literature is not research



Research is…

Searching for explanation of events, phenomena, relationships and causes

- What, how and why things occur
- Are there interactions?

A process

- Planned and managed –to make the information generated credible
- The process is creative
- It is **circular**–always leads to more questions

Research is…

All **well** designed and conducted research has potential application.

- Failure to see applications can be due to:
  - Users not trained or experienced in the specialized methods of economic research and reasoning
  - Researchers often do not provide adequate interpretations and guidance on applications of the research

Researchers are responsible to help users understand research implications

(How?)

What is Research?

"Research is the systematic approach to obtaining and confirming new and reliable knowledge"

–Systematic and orderly (following a series of steps)

–Purpose is **new** knowledge, which must be reliable

This is a general definition which applies to all disciplines

General Objectives of Research

- To gain **familiarity with phenomena and achieve new insight** into it.
- To portray accurately the **characteristic** of a particular individual, situation or a group.
- To determine the frequency with which something occurs or with which it is **associated** with something else.
- To discover **truth and fact.**
- To **understand** social life and gain a control over social behavior.
- To know old conclusions with new data.
- To find new conclusion with old data.
- To reach more conclusions from available data.
- To explain **unexplained horizon of knowledge**.
- To put forward an **entirely new theory**.
- To study and **resolve contradiction** in the area of a study.

Objectives of Research

In the field of **Computer Engineering**, the objectives of research are focused on advancing knowledge, developing innovative technologies, and solving practical and theoretical problems related to computation, software, and hardware.

Objectives:

- ✓ **Innovation**: Develop novel algorithms, architectures, and systems to enhance computational efficiency and capabilities.
- ✓ **Optimization**: Improve existing methods, technologies, or processes to make them faster, more efficient, or more secure.
- ✓ **Automation**: Design systems that can perform tasks with minimal human intervention.
- ✓ **Problem Solving**: Address real-world challenges in diverse areas like healthcare, finance, and education using computational methods.
- ✓ **Knowledge Advancement**: Contribute to theoretical foundations, such as complexity theory or data structures, to deepen understanding of computation.

Characteristics of Research

1. Empirical - based on observations and experimentation
2. Systematic - follows orderly and sequential procedure.
3. Controlled - all variables except those that are tested/experimented upon are kept constant. BOSSING
4. Employs hypothesis - process
5. Analytical - There is critical analysis of all data used so that there is no error in their interpretation
6. Objective, Unbiased, & Logical - all findings are logically based on empirical.
7. Employs quantitative or statistical methods - data are transformed into numerical measures and are treated statistically.



https://en.wikiversity.org/wiki/Characteristics_of_research

Your Research

Sources of research ideas

- Always be working on *something* (even if it won‟t be your thesis topic)
  - Think about problems, difficulties, or irritations
  - You might get sidetracked onto something more important
- Learn about other research: attend talks, read papers, take classes
  - future work they identified
  - connections they did not make
  - assumptions
  - other criticisms
- Your advisor
  - OK for starter projects, master's projects
  - not OK for thesis topic

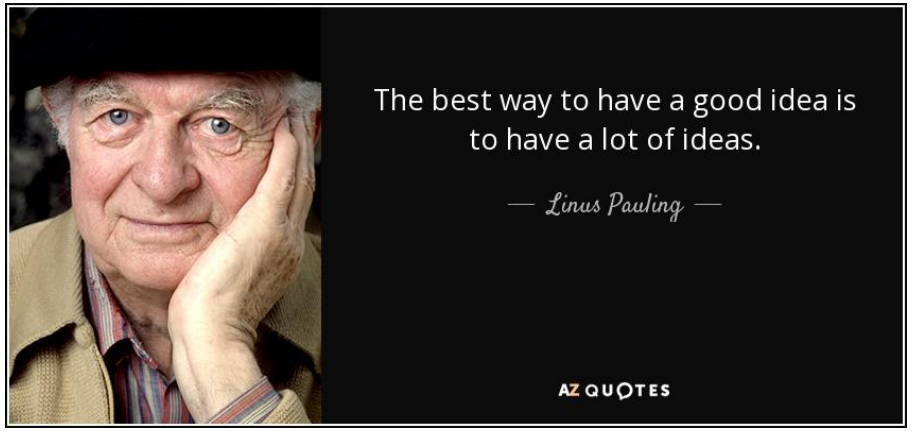
Your Research

- Think outside the box: A good research idea often "cheats"
  - If you take the same approach as other people, you are likely to be no more successful than they are
  - Cut the Gordian knot: take a different approach

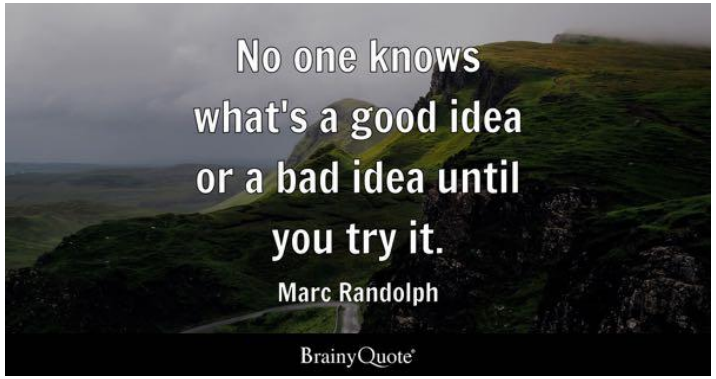


Your Research

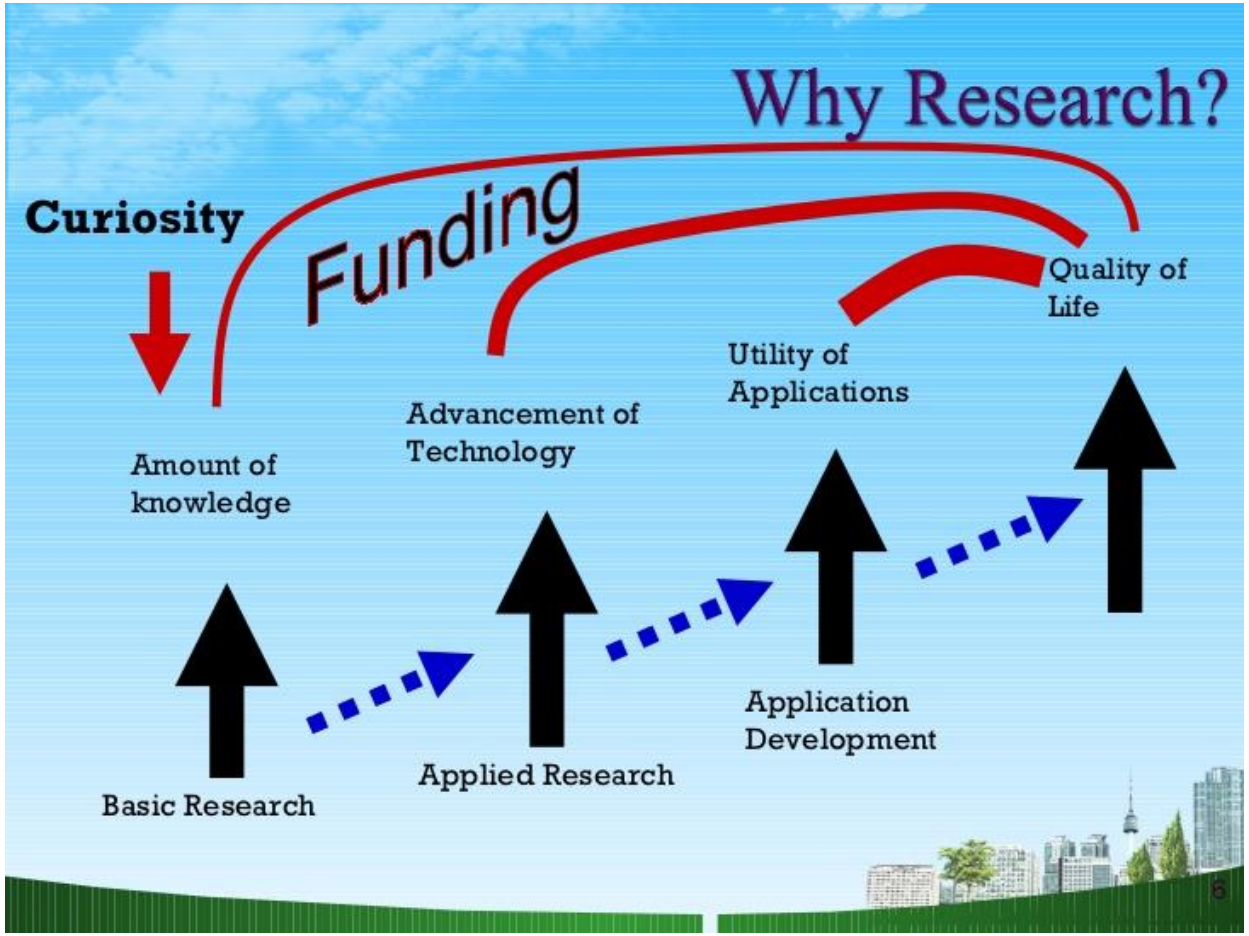
- Bad ideas are essential
- Most ideas are bad ideas
- If you have lots of ideas, some will be good
- It's better to have bad ideas than no ideas
  - I can teach people how to filter
  - It's harder to teach courage and imagination



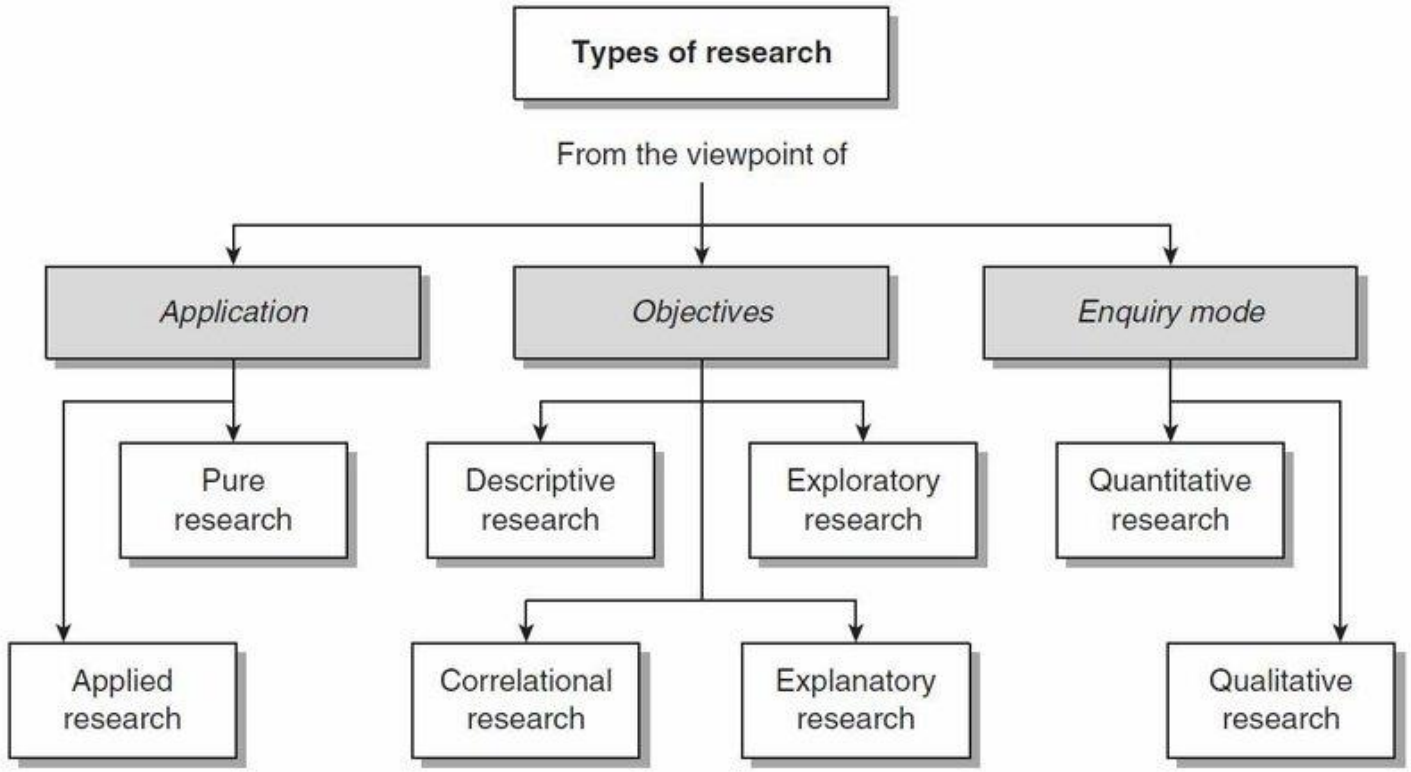
- Filtering is initially your advisor's role
  - You will get better with time, and spend less time evaluating and discarding bad ideas
  - Have faith in your own judgment, too
  - Your advisor is not foolproof.
- Generate lots of ideas before filtering
  - Breadth-first, not depth-first, approach to the problem



Objectives of Research



Types of Research



(Kumar, 2011)

Disciplinary, Subject-matter, and Problem-solving Research (Johnson, 1986)

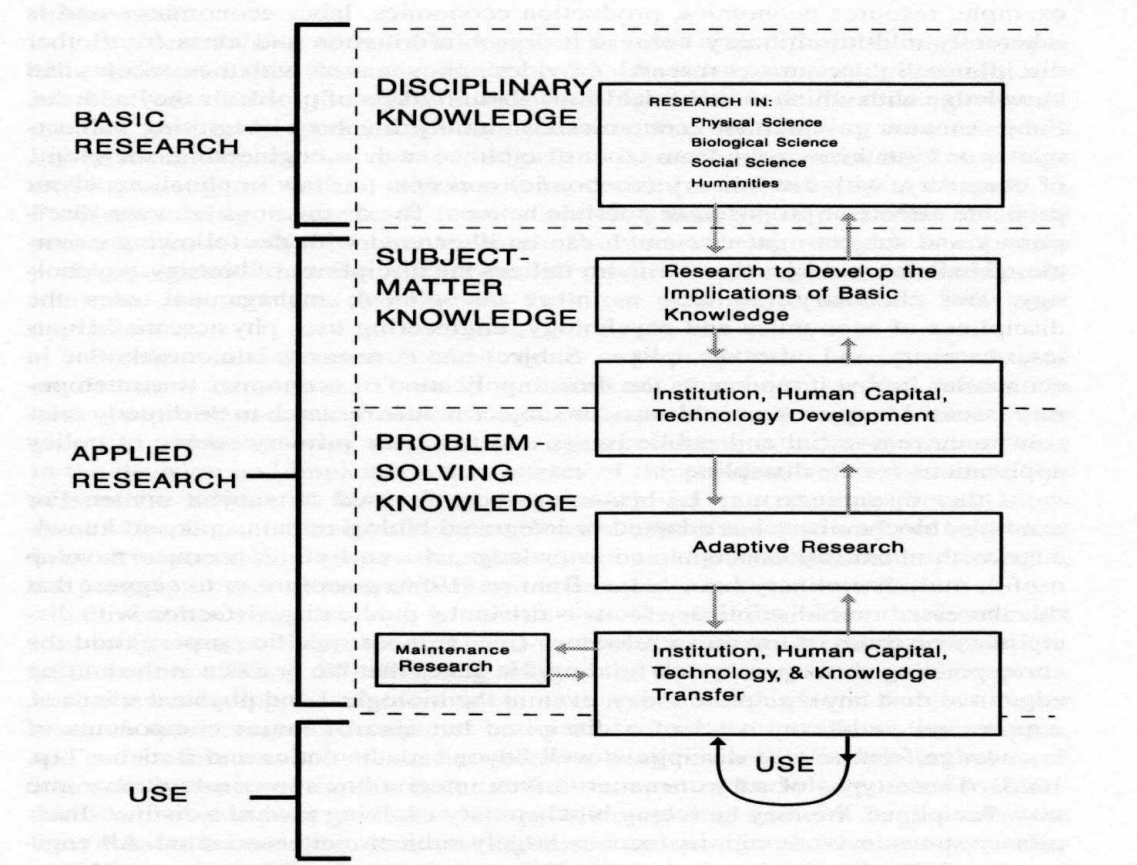


Figure 2.1. Relationship among research, knowledge, and use (Bonnen, 1986).

Analytic vs Descriptive Research

Descriptive Research–the attempt to determine, describe, or identify something

- The intent is often synthesis, which pulls knowledge or information together

Analytic–the attempt to establish why something occurs or how it came to be

- All disciplines generally engage in both

Based on Purpose

Basic (or Fundamental) Research

- expanding knowledge without immediate practical application
- Focuses on understanding underlying principles and theories

Applied Research

- Focused on solving specific, practical problems
- Uses findings from basic research for real-world applications

Exploratory Research

- Conducted to explore a new or poorly understood topic
- Aims to identify patterns, ideas, or hypotheses for future studies.
- Example: Investigating how people use a 5G technology.

Descriptive Research

- Seeks to describe characteristics or phenomena systematically
- Often involves surveys, case studies, or observational studies

Explanatory (or Causal) Research

- Aims to determine cause-and-effect relationships.
- Often uses experiments or longitudinal studies.
- Example: Examining the impact of a ICT training program on employee productivity.

Based on Methodology

Qualitative Research

- Focuses on exploring ideas and understanding experiences.
- Relies on non-numeric data like interviews, focus groups, and text analysis.
- Example: Understanding customer perceptions of a brand.

Quantitative Research

- Involves numeric data and statistical analysis to test hypotheses.
- Example: Measuring the effectiveness of an ad campaign through surveys.

Mixed-Methods Research

- Combines qualitative and quantitative approaches to provide a comprehensive understanding.
- Example: Studying Automation System satisfaction through surveys (quantitative) and interviews (qualitative).

Quantitative Vs Qualitative

A quantitative approach is one in which the investigator primarily uses post-positivist claims for developing knowledge (i.e. cause and effect thinking, reduction to specific variables and hypotheses and questions, use of measurement and observation, and the test of theories). (Creswell, 2003, p.19)

… qualitative researchers study things in their natural settings, attempting to make sense of or interpret phenomenon in terms of the meanings people bring to them. (Denzin& Lincoln, 2000, p.3).

Based on Time Frame

a. Cross-Sectional Research

- Studies phenomena at a single point in time.
- Example: A survey on public opinion about 5G.

b. Longitudinal Research

- Observes the same subjects over an extended period.
- Example: Tracking the academic performance of students over 10 years.

Based on Data Source

a. Primary Research

- Involves the collection of new data directly from the source.
- Example: Conducting experiments or surveys.

b. Secondary Research

- Analyzes existing data collected by others.
- Example: Reviewing census data to study population trends.

Based on Discipline

a. Social Science Research

- Focuses on studying human behavior and societies.
- Example: Research on cultural norms in different countries.

b. Natural Science Research

- Investigates the natural world, including biology, physics, and chemistry.
- Example: Studying the effects of climate change on marine ecosystems.

c. Interdisciplinary Research

- Combines insights from multiple disciplines to solve complex problems.
- Example: Researching sustainable urban development using environmental science and urban planning.

Specialized Types

a. Action Research

- Focuses on solving immediate problems within a specific context, often involving collaboration with stakeholders.
- Example: Improving teaching methods in a classroom.

b. Experimental Research

- Manipulates variables to study cause-and-effect relationships.
- Example: Testing the efficacy of a new solution.

c. Case Study Research

- In-depth analysis of a single instance, event, or entity.
- Example: Studying the rise of a IT successful startup.

d. Ethnographic Research

- Immerses the researcher in a culture or community to understand it deeply.
- Example: Studying the daily lives of indigenous people.

6P's of Research

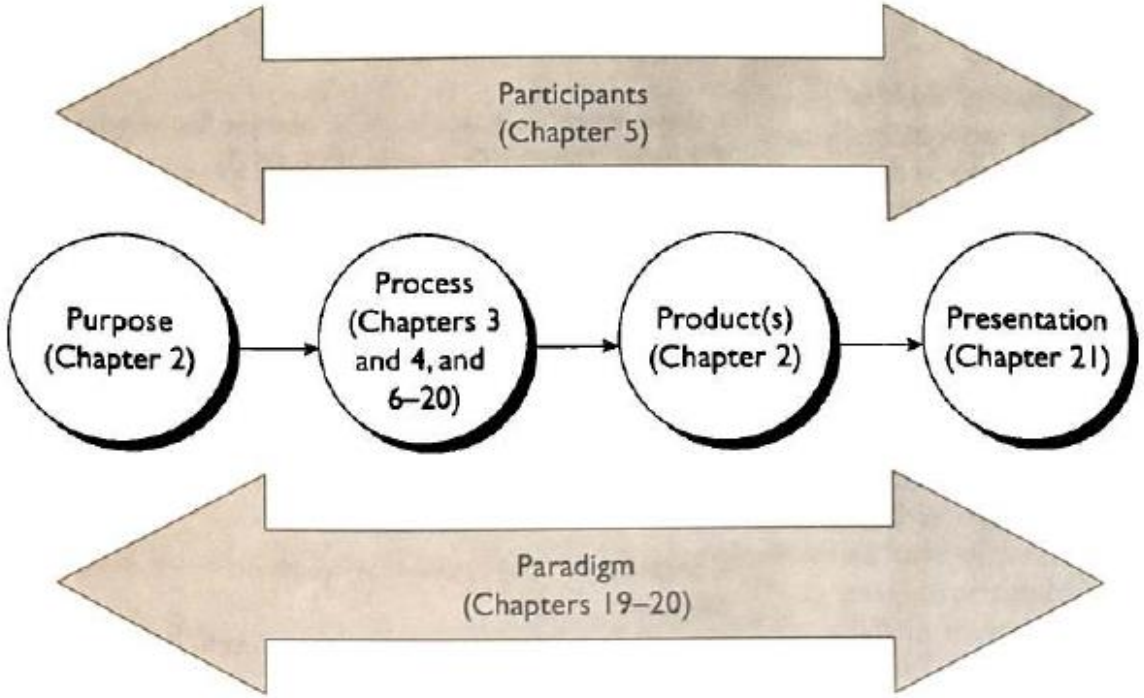


Figure 1.4 The 6Ps of research

Purpose

✓The reason for doing the research, the topic of interest, why it is important or useful to study this, the specific research question(s) asked and the objectives set. Research without a purpose is unlikely to be good research.

- What is the research problem under investigation?
- What are the reasons for doing the research?
- What are the objectives (more general) and aims (more specific) of the research?
- Is the research relevant to the field of study and to other people?

Products

The outcomes of research, especially your contribution to knowledge about your subject area. Your contribution can be an answer to your original research question(s) but can also include unexpected findings. For example, you and the academic community might learn something about a particular research strategy as a result of your research. Your thesis, dissertation, conference paper or journal article is also a product of your research.

- What are the expected and unexpected outcomes of the research?
- What is the contribution of knowledge to the subject of study, field of interest or research community?
- What aspects of the research need further investigation?

Process

The sequence of activities undertaken in any research project. The process involves identifying one or more research topics, establishing a conceptual framework (the way you choose to think about your research topic), the selection and use of a research strategy and data generation methods, the analysis of data and the drawing of conclusions, including recognizing any limitations in your own research.

- What is the format and layout of the conceptual framework of the researcher?
- How was the research conducted?
- What is the research strategy?
- What data collection methods were used?
- How was the data analyzed?
- What conclusions were drawn?
- Was the research process done in a systematic way and is it valid research process?
- What are the boundaries of the research?
- What are the limitations of the research?

Participants

These include those whom you directly involve in your research, for example by interviewing them or observing them, and also those who are indirectly involved, such as the editors to whom you submit a research paper. It is important that you deal with all these people legally and ethically, that is, you do not do anything that might annoy them or cause them harm (physically, mentally or socially).

- How is the researcher involved in the research?
- What are the legal and ethical implications of the research?
- Consideration and respect for all people directly or indirectly involved in the research.

Paradigm

A pattern or model or shared way of thinking. Managers sometimes talk of the need for a 'paradigm shift' to mean that a new way of thinking is required.

Example: object-oriented paradigm (for example C++)

Philosophical paradigms:

- Positivism- positivism asserts that knowledge is derived from observable phenomena and measurable facts. Emphasis on objectivity, replicability, and reliability.
- Interpretivism- Emphasizes understanding the subjective meaning of human experiences. Reality is seen as socially constructed and context-dependent.
- Critical Research- Based on critical theory, it challenges societal structures and power dynamics.

What is the underlying philosophical research paradigm assumed?

Presentation

The means by which the research is disseminated and explained to others.

For example, it may be written up in a paper or thesis, or a conference paper is presented to an audience of conference delegates, or a computer-based product is demonstrated to clients, users or examiners.

It is important that the presentation is carried out professionally — otherwise your audience might assume your whole research project was not undertaken in a professional manner.

- How is the research communicated to others who are interested?
- Is the research professionally conveyed by means of a dissertation, thesis, conference contribution, article or product?

The Research Process

1. Identification of general problem/question
2. Literature review
3. Specify questions/hypotheses
4. Determination of design/methodology
5. Data collection
6. Data analysis/presentation
7. Interpretation of findings

Research Process

- The process is initiated with a **question or problem** (**step 1**)
- Next, goals and **objectives are formulated** to deal with the question or problem (**step 2**)
- Then the **research design** is developed to achieve the objectives (**step 3**)
- **Results** are generated by conducting the research (**step 4**)
- **Interpretation** and analysis of results follow (**step 5)**

Creativity in the Research Process

- Research is a creative process
- *"…research includes far more than mere logic … It includes insight, genius, groping, pondering – „sense" … The logic we can teach; the art we cannot" (p 30)*
- Research requires (or at least works best) with imagination, initiative, intuition, and curiosity.
- There are different types of creativity, characteristic of different situations –"applied" and "theoretical" most closely associate with research
- 15

Fostering Creativity (Ladd 1987)

A. Gather and use previously developed *knowledge*

B. Exchange ideas

C. Apply deductive logic

D. Look at things alternate ways

E. Question or challenge assumptions

F. Search for patterns or relationships

G. Take risks

H. Cultivate tolerance for uncertainty

I. Allow curiosity to grow

J. Set problems aside … and come back to them

K. Write down your thoughts

"… frequently I don't know what I think until I write it"

L. Freedom from distraction … some time to think.

Creativity may provide the difference between satisfactory and outstanding research.

Research in Computer Science and Engineering

Biomedical Informatics

Biomed Imaging and Visualization

Cloud Computing

Cybersecurity

Cyber-Physical Systems

Databases and Data Mining

Data Science and Analytics

Multimedia Systems and Apps

Semantic, Social and Sensor Web

Machine Learning and Artificial Intelligence

Wireless Networking and Security

https://cse.engin.umich.edu/research/research-areas/

Research in **Computer Science and Engineering (CSE)** covers a broad range of areas aimed at advancing computing technologies, addressing societal challenges, and enabling innovative solutions. Key research areas include:

1. Artificial Intelligence and Machine Learning (AI/ML)

- **Topics**: Deep learning, reinforcement learning, natural language processing, computer vision, generative AI, ethical AI.
- **Applications**: Healthcare, autonomous systems, smart cities, and predictive analytics.

2. Data Science and Big Data

- **Topics**: Data mining, data visualization, large-scale data storage, and real-time analytics.
- **Applications**: Business intelligence, epidemiology, climate modeling, and sentiment analysis.

3. Cybersecurity and Privacy

- **Topics**: Cryptography, threat modeling, secure protocols, and quantum-safe algorithms.
- **Applications**: Financial systems, IoT security, digital identity protection.

4. Human-Computer Interaction (HCI)

- **Topics**: Usability engineering, augmented/virtual reality interfaces, and accessibility technologies.
- **Applications**: Gaming, assistive technologies, and immersive learning environments.

5. Internet of Things (IoT)

- **Topics**: Sensor networks, edge computing, and IoT security.
- **Applications**: Smart homes, healthcare monitoring, and industrial automation.

6. Networking and Communications

- **Topics**: 5G/6G technologies, software-defined networking, and satellite communications.
- **Applications**: Telecommunication infrastructure, disaster recovery networks, and rural connectivity.

7. Software Engineering

- **Topics**: Agile methodologies, DevOps, formal methods, and software verification.
- **Applications**: Large-scale systems development, enterprise applications, and fault-tolerant software.

8. Computer Vision and Image Processing

- **Topics**: Object recognition, video analytics, and biometrics.
- **Applications**: Autonomous vehicles, surveillance, and medical imaging.

9. Quantum Computing

- **Topics**: Quantum algorithms, quantum cryptography, and error correction.
- **Applications**: Optimization, complex simulations, and secure communications.

10. Cloud Computing and Edge Computing

- **Topics**: Distributed systems, multi-cloud architectures, and green computing.
- **Applications**: Scalable applications, real-time processing, and cost-efficient storage.

11. Robotics and Autonomous Systems

- **Topics**: Path planning, swarm intelligence, and human-robot collaboration.
- **Applications**: Manufacturing, agriculture, and exploration.

12. Theoretical Computer Science

- **Topics**: Computational complexity, graph theory, and automata theory.
- **Applications**: Algorithm optimization, database indexing, and machine learning foundations.

13. Bioinformatics and Computational Biology

- **Topics**: Genome sequencing, protein structure prediction, and systems biology.
- **Applications**: Drug discovery, personalized medicine, and evolutionary studies.

14. Green and Sustainable Computing

- **Topics**: Energy-efficient algorithms, carbon-aware systems, and circular computing.
- **Applications**: Climate monitoring, energy conservation, and e-waste management.

15. Virtual Reality (VR) and Augmented Reality (AR)

- **Topics**: Real-time rendering, haptic feedback, and spatial computing.
- **Applications**: Gaming, training simulations, and medical imaging.

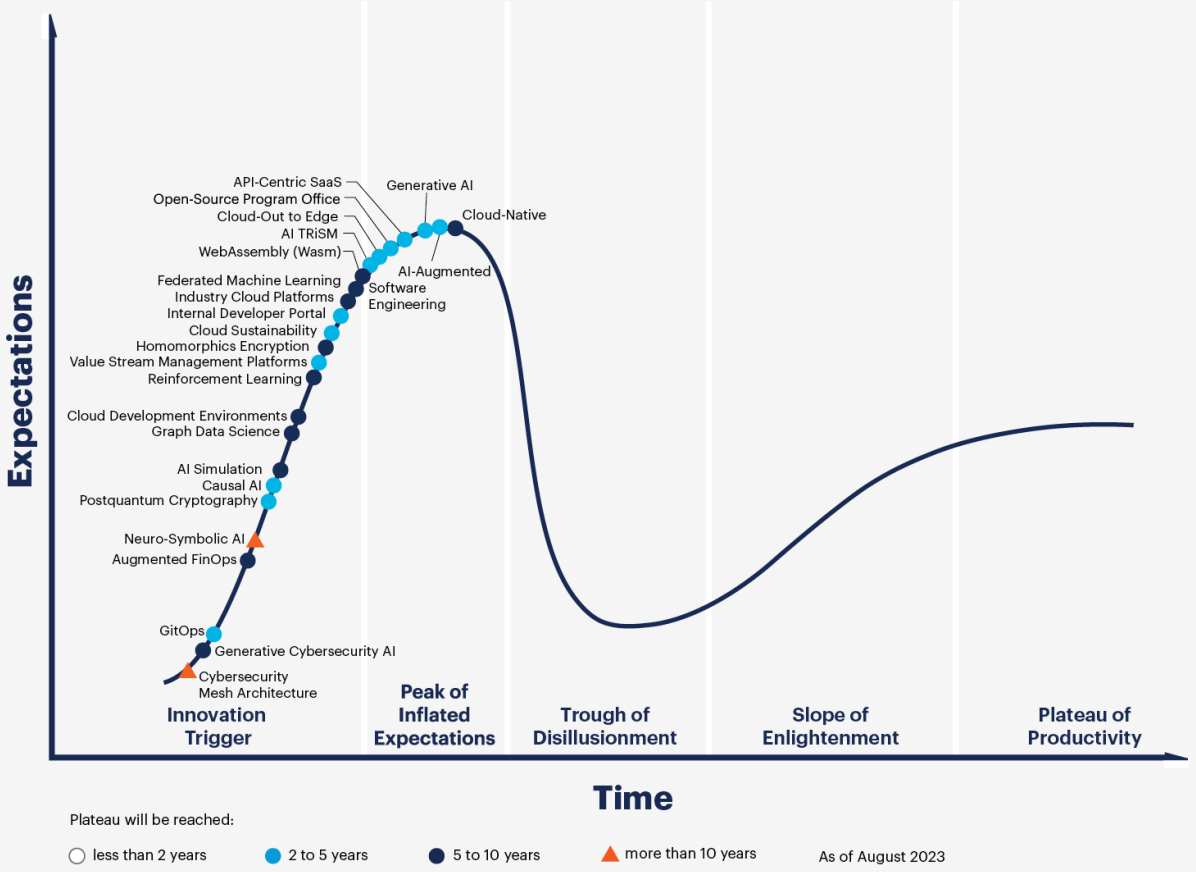
Approaches to Research

- **Interdisciplinary Research**: Collaboration with fields like biology, economics, or psychology.
- **Problem-Driven Innovation**: Focusing on solving specific societal or industrial challenges.
- **Emerging Trends**: Areas like digital twins, synthetic data, or AI ethics are gaining traction.

Tips for Effective Research in CSE

- **Stay Updated**: Follow top conferences like NeurIPS, ACM SIGCOMM, or IEEE CVPR.
- **Collaboration**: Engage in multidisciplinary projects.
- **Publication**: Publish in reputable journals and conferences.
- **Practical Applications**: Aim for research with measurable real-world impact.

Hype Cycle for Emerging Technologies, 2023

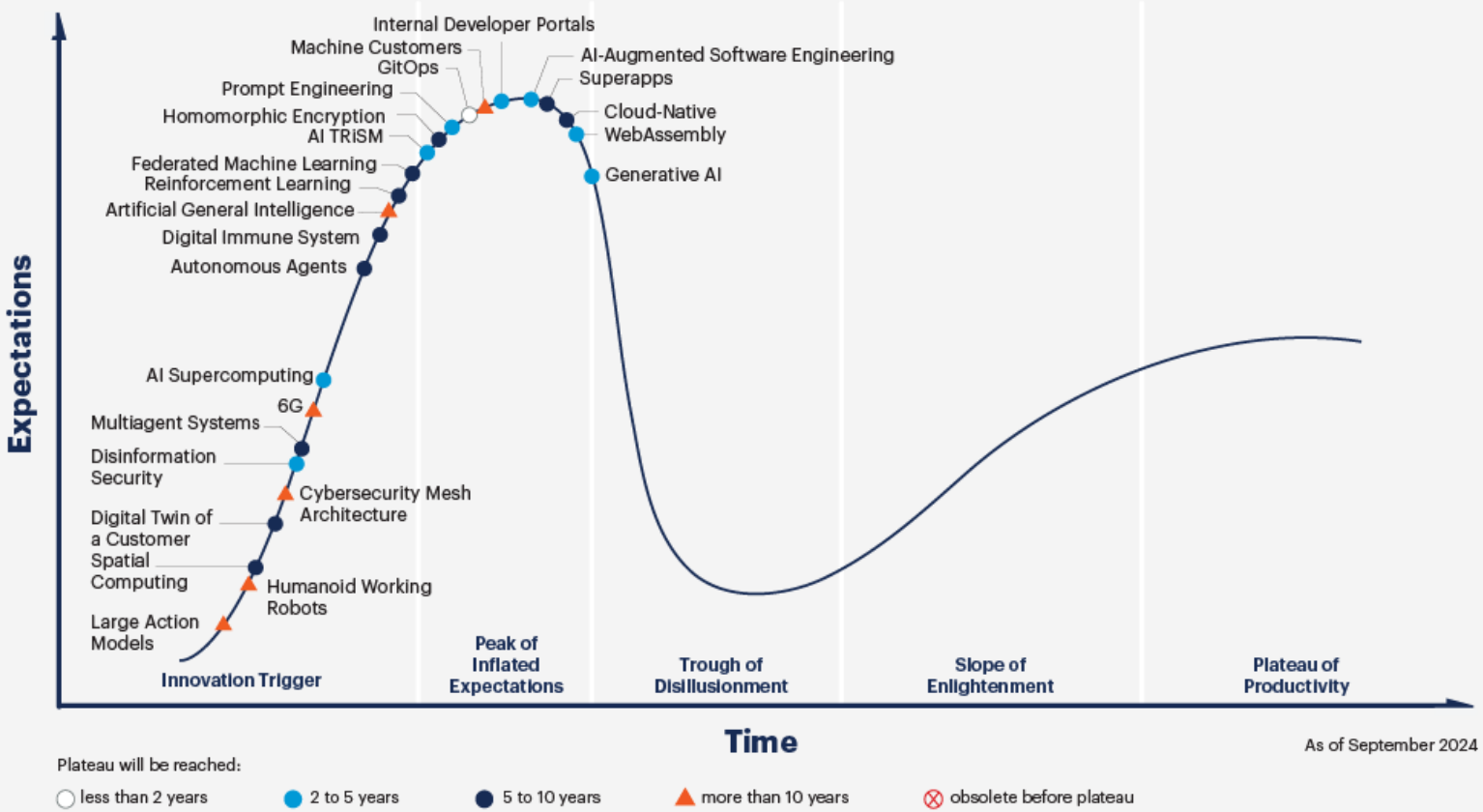


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Hype Cycle for Emerging Technologies, 2024



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